

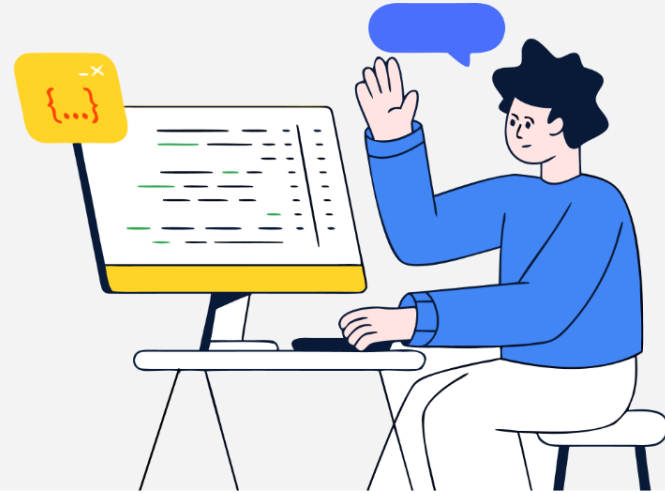
Guidelines

- Kindly use the given template for submitting your updated prototype(Make a copy of the template)
- One team is only required to submit one project.
- Follow this template while preparing your presentation.
- You are welcome to add as many POCs and design concepts to support your project.
- In case of queries, kindly reach out to us at solutionchallengesupport@hack2skill.com

Note: Updated Prototype must include at least one Google AI model or service (e.g., Gemini) as part of the solution.



Solution Challenge



Team Details

- Team name: FairFlow AI**
- Team leader name: Aaditya Kothari**
- Problem Statement: AI systems used in hiring, lending, and healthcare in India perpetuate bias against protected groups including gender, caste, religion, and regional origin — with no accessible tool to detect, explain, or mitigate this discrimination.**

FairFlow AI is an end-to-end AI fairness governance platform built for India.

What it does:

Audits AI decision models (hiring, lending, healthcare) for bias across gender, caste, religion, and regional origin.

Key additions in this prototype:

Client-side WASM privacy shield — fairness pre-check computed in the browser before any data is uploaded.

LangGraph agentic audit pipeline with Observe → Analyze → Act → Report state machine.

DoWhy causal proxy detection — identifies variables that act as proxies for protected attributes.

Differential privacy with Laplace noise for all audit certificates.

India Context Scan — caste, religion, and surname proxy detection with Article 15 and Article 16 compliance analysis via Gemini.

Job Description Bias Detector — proactive bias prevention before a JD is published.

Export to Google Docs — audit report exported directly to user's own Google Drive via OAuth.

Opportunities — Why FairFlow AI Stands Apart

How is it different from existing solutions?

IBM AIF360, Google What-If Tool, and Fairlearn are all desktop/API tools with no mobile access, no plain-language explanations, and zero India-specific bias detection. FairFlow is the only tool that audits for caste and religion bias with Indian constitutional law citations.

How does it solve the problem?

A recruiter uploads a hiring CSV. In under 60 seconds: WASM pre-check in browser, full ML audit on server, Gemini plain-language verdict citing Article 15, causal proxy map, mitigation comparison, and a downloadable compliance report. No data science expertise required.

USP of FairFlow AI

India caste/religion bias detection — unique in the global fairness tool market.

TCAV concept sensitivity analysis — research-grade explainability in a student project.

WASM privacy shield — zero egress pre-check, data stays in browser until user chooses to upload.

Differential privacy audit certificates — tamper-evident with SHA-256 hash.

SDG 10.3 — directly addresses Reduced Inequalities in AI-driven decisions.

Complete Feature List

Core Audit Engine

Disparate Impact, Statistical Parity, Equal Opportunity, Average Odds — 4 fairness metrics with PASS/FAIL thresholds.

SHAP explainability for individual candidate decisions.

Privacy and Security

WASM C++ SHA-256 browser-side pre-check with zero data egress.

Differential privacy (Laplace noise, configurable epsilon) on all audit results.

GDPR Right-to-Erasure endpoint — full cascade delete.

MIME magic-byte file validation.

India-Specific

Caste, religion, and surname proxy detection.

Gemini analysis citing Article 15, Article 16, SC/ST Act 1989, RBI Fair Practice Code.

Advanced ML

DoWhy causal DAG with backdoor estimation — identifies proxy discrimination paths

Process Flow — FairFlow AI Audit Pipeline

1. User uploads CSV dataset via React frontend
2. WASM engine runs client-side pre-check (DI, proxy score) — zero data egress
3. User reviews pre-check results and confirms upload
4. FastAPI backend receives file — MIME magic-byte validation, rate limiting
5. ML pipeline: Disparate Impact, Statistical Parity, Equal Opportunity, Average Odds computed
6. DoWhy causal DAG built — proxy variables identified with correlation and causal effect scores
7. India Context Scan — surname proxy, caste/religion inference, regional bias flags
8. TCAV concept sensitivity — EducationPrestige, OperationalStability, Leadership scores
9. SHAP individual explanations computed for each candidate
10. LangGraph agent (Observe→Analyze→Act→Report) generates governance recommendation
11. Gemini 2.0 Flash generates plain-language verdict citing Article 15, Article 16, domain laws
12. Differential privacy applied — Laplace noise, SHA-256 certificate hash generated
13. Results displayed: Fairness Snapshot, India Scan, TCAV, Governance page
14. User runs Mitigation: Reweighting / Prejudice Remover / Equalized Odds compared
15. Gemini Executive Action Plan generated based on mitigation results
16. Export: PDF download or Google Docs (user OAuth — doc in their Drive)

UI Wireframes — Key Screens

Screen 1: Upload Audit

File picker → WASM pre-check card appears instantly (DI score, proxy feature, 0 data sent) → Upload button → Processing spinner

Screen 2: Audit Results

WASM Precheck card (FLAG/PASS metrics) → Fairness Snapshot (4 metric cards with FAIL/PASS) → Gemini Analysis (Article 15/16 citations, High Risk badge) → India Context Scan (caste/religion score, probabilistic disclaimer)

Screen 3: Mitigation Center

TCAV concept influence bars → Ethos Agent Recommendation → Strategy comparison chart (Before/Reweighting/Equalized Odds) → Fairness score 50→75 arrow → Gemini Executive Action Plan → Export buttons

Screen 4: Governance Intelligence

Governance Verdict FLAG card → Causal Proxy Risk Panel (zip_code corr=1.000, education_level, ethnicity, age) → DoWhy Path View (gender→hired) → Memory Timeline (LangGraph audit history) → Fairness Glossary

Screen 5: JD Bias Checker

Architecture — FairFlow AI

Frontend Layer (React.js + Firebase Hosting)

React pages: Dashboard, Upload Audit, Candidates, Mitigation, Governance, JD Bias Checker

WASM Module: C++ SHA-256 + fairness pre-check compiled to WebAssembly via Emscripten

Firebase Auth: Google OAuth with Drive and Docs scopes

Backend Layer (FastAPI on Google Cloud Run)

Routers: /audit, /mitigate, /governance, /jd-audit, /candidates, /report, /privacy

ML Core: bias_detector.py, mitigator.py, causal_tcav.py, counterfactual.py, explainer.py, india_bias_detector.py, cultural_audit.py, synthetic_patch.py

LangGraph Agent: auditor_graph.py (TypedDict state machine) + memory_store.py (HashingVectorizer RAG)

Privacy: differential_privacy.py (Laplace noise), GDPR delete endpoint, SHA-256 certificate

AI Services (Google)

Gemini 2.0 Flash: audit verdicts, India law citations, JD alternatives, mitigation plans

Google Docs API + Drive API: OAuth report export to user's Drive

Data Layer

PostgreSQL + SQLAlchemy ORM — audit records, user accounts, GDPR-compliant cascade delete

Technologies Used

Google AI and Cloud

Gemini 2.0 Flash — plain-language bias verdicts, India law citations, JD bias alternatives, mitigation action plans.

Firebase Auth — Google OAuth for user login and Google Docs export.

Google Docs API and Drive API — audit report export to user's Drive.

ML and Fairness

scikit-learn, AIF360, DoWhy, SHAP, LangGraph, HashingVectorizer for local RAG.

Privacy and Security

WebAssembly (C++ compiled to WASM) — browser-side SHA-256 and fairness pre-check.

Differential Privacy — Laplace mechanism with configurable epsilon.

Backend and Frontend

FastAPI (Python 3.13), PostgreSQL, SQLAlchemy, Docker, React.js, Recharts, Tailwind CSS.

Estimated Implementation Cost

Monthly Cloud Costs (Estimated)

Google Cloud Run — \$0 (free tier, scales to ~\$15/month at 10k requests).

Firebase (Auth + Hosting) — \$0 free tier.

Gemini API — \$0.075 per 1M input tokens (Flash). Estimated \$5–15/month at moderate usage.

PostgreSQL (Cloud SQL) — ~\$10/month (db-f1-micro).

Google Docs API — free tier (300 requests/minute).

Total estimated monthly cost: \$15–30/month for a production MVP.

Development Cost

Built entirely by student team. Zero licensing fees. All ML libraries open source.

WASM compiled from open-source C++ SHA-256 implementation.

MVP Snapshots — FairFlow AI

The following pages show live screenshots of the working prototype:

1. WASM Privacy Shield: Client-side fairness pre-check — $DI=0.5385$ flagged before upload. Processed 200 rows with zero data egress.
2. Fairness Snapshot: 4-metric bias assessment — Disparate Impact FAIL (0.5385), Statistical Parity FAIL (-0.30), Equal Opportunity PASS, Average Odds PASS. Overall score 50/100.
3. Gemini AI Fairness Verdict: Plain-language analysis citing Article 15 and Article 16. High Risk badge. Disparate Impact 0.5385 — hired only 54% as often as others. Bottom line: Urgent action required.
4. India Context Scan: Caste/Religion/Regional bias. Score 80/100. Surname Proxy detected. Probabilistic methodology disclaimer shown.
5. Mitigation Center: TCAV concept sensitivity (EducationPrestige $tcav=0.46$, OperationalStability $tcav=0.51$). Ethos Agent recommending Equalized Odds. Fairness score 50→75 after Reweighting.
6. Governance Intelligence: Causal proxy panel — $zip_code\ corr=1.000$, DoWhy path $gender \rightarrow hired$. LangGraph memory timeline. Fairness glossary.
7. JD Bias Checker: Detected rockstar, ninja, wizard (male-coded), young, fast-paced (exclusionary). Gemini alternatives: high-performing professional, highly skilled specialist.
8. PDF Report and Google Docs export: Stakeholder-ready compliance report downloadable as PDF or exported to user's own Google Drive via OAuth.

Google Technologies Used

Gemini 2.0 Flash (Core AI)

Powers all plain-language bias verdicts, India constitutional law citations, JD bias word replacements, and mitigation action plans. The Gemini prompt explicitly references Article 15, Article 16, SC/ST Atrocities Act 1989, RBI Fair Practice Code, and Clinical Establishments Act 2010.

Firebase Authentication

Google Sign-In with Drive and Docs OAuth scopes. User token used for all Google Workspace operations — no service account, data never touches our infrastructure.

Google Docs API and Drive API

Audit reports exported directly to user's own Google Drive as a formatted Google Doc. User retains full ownership and control.

Project Links

GitHub Public Repository

<https://github.com/Akshat-coder2106/Fair-Flow>

Demo Video (3 Minutes)

<https://youtu.be/XnDJE2lyyvl>

Working Prototype (Live URL)

<https://fairflow-frontend.onrender.com>

Contact

Team FairFlow AI — SDG 10.3 — Reducing AI Discrimination in Hiring, Lending and Healthcare





Solution Challenge



Thank You